

Nanopublications for FAIR and AI-Ready Data: Building a Decentralized Knowledge Commons

Ashley Caselli^{1,2,*}, Tobias Kuhn^{1,3}, Virginia Balseiro¹ and Ziroli Plutschow¹

¹Knowledge Pixels, Switzerland

²University of Geneva, Switzerland

³Vrije Universiteit Amsterdam, Netherlands

Abstract

Knowledge sharing is a fundamental aspect of scientific research, but it is often made difficult by the lack of standardized formats and channels for sharing data. Challenges like data integration, interoperability, and provenance traceability are common obstacles faced by organizations and researchers.

To cope with this, we present a decentralized publishing platform that enables researchers and machines to share, discover, and reuse knowledge in a trusted and machine-actionable way. The platform is built on immutable and digitally signed RDF packages that encapsulate an assertion, its provenance, and publication metadata, known as nanopublications, which serve as the foundational building block enabling the platform's core functionalities. We briefly discuss the technical foundations of the platform and its impact on data governance, reproducibility, and AI-readiness. To illustrate the platform in practice, we mention four use cases developed within the Knowledge Pixels Incubator program, spanning chemical safety, open science communication, healthcare policy, and biotech provenance. Finally, we show how this platform can transform the research cycle into a machine-actionable knowledge commons providing high-precision data required for the next generation of AI tools.

Keywords

Nanopublications, FAIR Principles, Decentralized Knowledge Graphs, AI-Ready Data, Provenance, Data Governance

1. Introduction and Problem Statement

Research data is often shared in a non-machine-interpretable manner, often including off-the-record information that is not captured in a structured way, which makes it difficult to integrate and reuse. In addition, there is no global platform for sharing and accessing this information, which leads to silos and fragmentation of knowledge, making it a challenge for finding it and using it effectively. This is one of the major obstacles for the advancement of science, as it limits the ability of researchers to build upon existing reliable knowledge and to collaborate effectively. To address this, we propose a decentralized publishing platform designed to make any piece of structured knowledge permanently accessible, citable, and machine-actionable. The platform is built on nanopublications, self-contained and globally addressable knowledge graph snippets initially designed for capturing individual scientific claims [1, 2], and subsequently extended to represent arbitrary structured information [3, 4], as the enabling technology that allows the platform to operate in a decentralized, trustworthy, and interoperable manner. Each nanopublication encapsulates an assertion, its provenance, and publication metadata, and constitutes a node in the larger knowledge graph that the platform collectively maintains across a decentralized network of publishing, query, and application services.

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*Corresponding author.

✉ ashley@knowledgepixels.com (A. Caselli); tobias@knowledgepixels.com (T. Kuhn); virginia@knowledgepixels.com (V. Balseiro); ziroplu@gmail.com (Z. Plutschow)

🌐 <https://ashleycaselli.github.io/> (A. Caselli); <https://www.tkuhn.org/> (T. Kuhn); <https://virginiabalseiro.com/> (V. Balseiro)

🆔 0000-0001-8492-0354 (A. Caselli); 0000-0002-1267-0234 (T. Kuhn); 0009-0009-0118-9195 (V. Balseiro); 0009-0008-3635-347X (Z. Plutschow)



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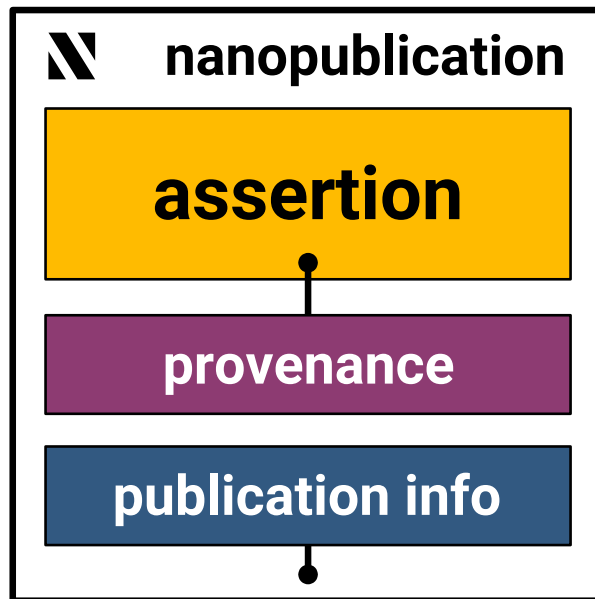


Figure 1: The structure of a nanopublication.

2. Technology and Methodology

This section describes the technical foundations of the decentralized publishing platform, with a focus on the role of nanopublications as the enabling technology and the architecture of the platform built around them. We first detail the structure of a nanopublication as a self-contained, verifiable unit of publication, and then examine how the platform as a whole aligns with the FAIR principles, demonstrating that FAIR compliance emerges from the design of the platform rather than being imposed as an external requirement.

2.1. Nanopublication Technology

As briefly mentioned in Section 1, a nanopublication is a logically self-contained and globally addressable RDF dataset that is treated as an independent publication and constitutes a node in a larger decentralized knowledge graph. Figure 1 depicts its structure, with more details that can be found in https://nanopub.net/guidelines/working_draft/.

It is composed of three named graphs: the *assertion* graph; the *provenance* graph; and the *publication info* graph. The assertion graph contains the main claim/s of the nanopublication, therefore where the actual knowledge can be found. The provenance graph contains metadata about the assertion, for instance who made the assertion or from where it was derived. Finally, the publication info graph contains metadata about the nanopublication itself, such as its creator, creation date, and license.

Each nanopublication is referred to using content-based immutable identifiers called Trusty URIs [5]. Such URIs ensure that the data is trustworthy and that traceability and provenance are maintained. A nanopublication can only be published after it is signed with a digital signature by a user. The signature as well as the signer identity and its public key are included in the publication info graph. This means that once a nanopublication is published, it cannot be altered without invalidating the signature, and anyone can verify the authenticity of the nanopublication by checking its signature.

The nanopublication platform is built on a three-layer peer-to-peer (P2P) network. Its open nature allows any party to deploy their own services and join the trusted network, ensuring that the broader ecosystem can grow and remain sustainable without dependence on any single provider. A comprehensive description and the technical documentation of the platform can be found at <https://nanopub.net/docs/architecture>.

The platform comprises publishing services, query services, and user-facing applications. The pub-

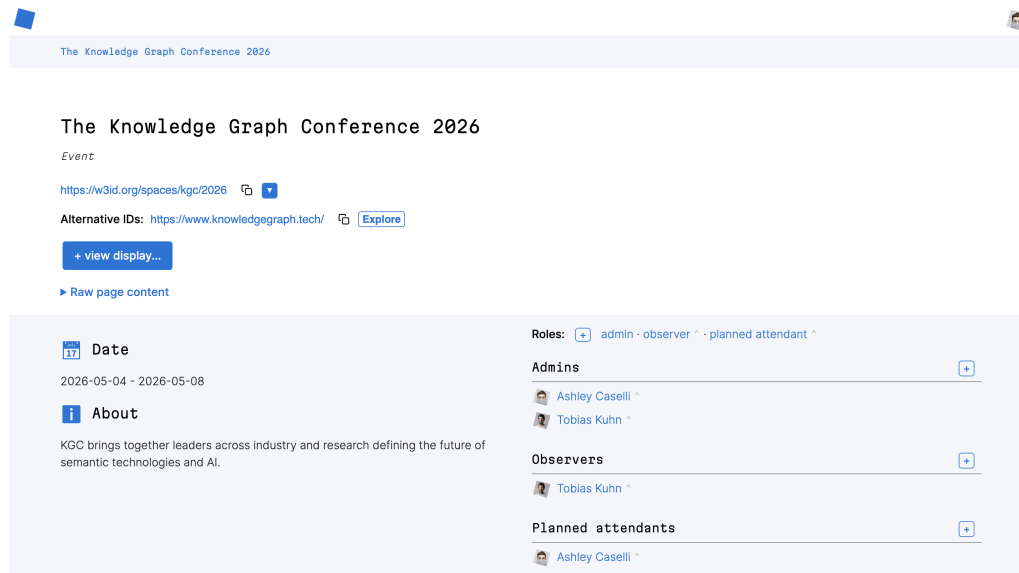


Figure 2: An example Space page in Nanodash for the “Knowledge Graph Conference 2026” context.

lishing layer provides nanopublication registries for storing and retrieving nanopublications; the query layer enables decentralized, global querying across the network; and the application layer offers user-friendly clients, such as Nanodash (<https://github.com/knowledgepixels/nanodash>), through which researchers can publish, browse, and query the knowledge commons directly. A typical publishing workflow using Nanodash consists of filling in a nanopublication template and publishing the resulting nanopublication to the network. During this process, quite often the user will want to link the nanopublication to a specific context, for instance a research project, a conference, or a community. This is supported by the platform through the concept of Spaces (<https://knowledgepixels.com/spaces/>), which are global representations for all kinds of collaborative units, and can be used to group and organize nanopublications within a specific context. Figure 2 shows an example of such a page in Nanodash, that defines a Space (of type Event) for the “Knowledge Graph Conference 2026” context.

Figure 3 illustrates how a template appears in Nanodash, with input fields for the user to fill in, and a submit button to publish the resulting nanopublication to the network. In this example, the template is designed to capture a “plan to attend” statement to a given event, as for instance the Knowledge Graph Conference 2026 event. Listing 1 shows the raw RDF of the resulting nanopublication, with its assertion graph containing the user’s “plan to attend” statement, and the provenance and publication info graphs containing the metadata about the assertion and the nanopublication itself, respectively. A more detailed overview of Nanodash and the publishing workflow in action can be found at https://www.youtube.com/watch?v=exJ_8p584cE.

2.2. FAIR Compliance

As shown in Section 2.1, the platform enables the creation and publication of nanopublications through structured templates, producing machine-actionable metadata on digital objects using open software. Trusty URIs are persistent identifiers (PIDs) that make each nanopublication findable and enable stable, verifiable references to the described object. In that sense, nanopublication templates are analogous to FAIR metadata schemas [6], as both provide a structured mechanism for attaching metadata and a PID to the object one wants to FAIRify.

Taken together, the platform offers a flexible and vendor-neutral pathway to FAIRification: it relies exclusively on open standards and software, and is designed to operate on top of existing IT systems and workflows without disrupting them.

```

@prefix this: <https://w3id.org/np/RA8xbP5i9BoadDAYnUim6YFiA6Q5l1VBn9UzjZtm0JUYw>
↪ .
@prefix sub: <https://w3id.org/np/RA8xbP5i9BoadDAYnUim6YFiA6Q5l1VBn9UzjZtm0JUYw/>
↪ .
@prefix np: <http://www.nanopub.org/nschema#> .
@prefix dct: <http://purl.org/dc/terms/> .
@prefix nt: <https://w3id.org/np/o/ntemplate/> .
@prefix npx: <http://purl.org/nanopub/x/> .
@prefix xsd: <http://www.w3.org/2001/XMLSchema#> .
@prefix rdfs: <http://www.w3.org/2000/01/rdf-schema#> .
@prefix orcid: <https://orcid.org/> .
@prefix prov: <http://www.w3.org/ns/prov#> .
@prefix foaf: <http://xmlns.com/foaf/0.1/> .

sub:Head {
  this: a np:Nanopublication;
  np:hasAssertion sub:assertion;
  np:hasProvenance sub:provenance;
  np:hasPublicationInfo sub:pubinfo .
}

sub:assertion {
  orcid:0000-0001-8492-0354 <https://w3id.org/kpxl/gen/terms/plansToAttend>
  ↪ <https://w3id.org/spaces/kgc/2026> .
}

sub:provenance {
  sub:assertion prov:wasAttributedTo orcid:0000-0001-8492-0354 .
}

sub:pubinfo {
  orcid:0000-0001-8492-0354 foaf:name "Ashley Caselli" .

  this: dct:created "2026-04-08T09:15:44.776Z"^^xsd:dateTime;
  dct:creator orcid:0000-0001-8492-0354;
  dct:license <https://creativecommons.org/publicdomain/zero/1.0/>;
  npx:hasNanopubType <https://w3id.org/kpxl/gen/terms/plansToAttend>;
  npx:wasCreatedAt <https://nanodash.knowledgepixels.com/>;
  rdfs:label "Plan to attend: The Knowledge Graph Conference 2026";
  nt:wasCreatedFromProvenanceTemplate
  ↪ <https://w3id.org/np/RA7lSq6MuK_TIC6JMSHvLtee3lpLoZD0qLJCLXevnrPoU>;
  nt:wasCreatedFromPubinfoTemplate
  ↪ <https://w3id.org/np/RACJ58Gvyn91LqCKIO9zu1eijDQIEff28iyDrJgJSJF8>,
  <https://w3id.org/np/RAukAcWHRDlkqkx7H2XNSegc1WnHI569INvNr-xdptDGI>;
  nt:wasCreatedFromTemplate
  ↪ <https://w3id.org/np/RAQ1LrU5fyY0Szd-4i9nN_JXBMvKti_v-GeHZgQG-EpzI> .

  sub:sig npx:hasAlgorithm "RSA";
  npx:hasPublicKey
  ↪ "MIGfMA0GCSqGSIB3DQEBAQUAA4GNADCBiQKBgQCaQXE2B2hJ6QdWxaWerU4F/XOtCQdj2OkdJ
  ↪ rIYd1kpMY9hIUh8boxa03uOBeqvqKlF5HjpRF4DT1eghOSsa2Dhd5HRTM/XIamvWkHpN4oBj
  ↪ 3Ml19pAAzjTqczYmFLfgG67mGL5+vSA8VEDHtKQCrvP4SpTibMBP1qaxt3k1tqD6wIDAQAB";
  npx:hasSignature "bxCHRoIOOgSffMa3dLUOFKa7ox7GWwNmRtzjGspSIO2Yh4MZMmwcPIXsIwJ
  ↪ HN+e0CC7d+JF+eK2zKmPdOuXGRXM3fd+7nYOpVEcc3tEdeZfG7e/SxL4u79w6IFZfjDhqY/dpJ
  ↪ s1IU043pyuV9h5Ym5hQesezbY04vL6NjsIT83/+8=";
  npx:hasSignatureTarget this:;
  npx:signedBy orcid:0000-0001-8492-0354 .
}

```

Listing 1: The nanopublication produced from the template in Figure 3, serialized in TriG format.

Create Nanopublication

Declaring to plan to attend an event [▲]

Such a nanopublication declares that the creator plans to attend an event.

Ashley Caselli plans to attend

The assertion above is attributed to

This nanopublication is created by

This nanopublication is published under the license

[show more](#)

☐ I understand that published data cannot be fully removed (only retracted or superseded by new versions), and is publicly connected to my personal identifier.

[Publish](#) [Preview](#)

Figure 3: A nanopublication template as it appears in the Nanodash client.

3. The Knowledge Pixels Incubator

The Knowledge Pixels Incubator is a program where new use cases around nanopublications can start and grow. During this program, we have been working with different partners to apply nanopublication-based approaches to address concrete knowledge-sharing challenges across diverse domains. More detailed information can be found in <https://w3id.org/spaces/knowledgepixels/incubator>.

Some of the use cases currently under development within the program are: *(i)* Chemical Safety: supporting a community of European institutes in collectively managing dynamic vocabularies for chemical risk assessment in a transparent and interoperable way; *(ii)* Open Science: addressing the loss of valuable contextual information in open science communication through structured, machine-readable nanopublications; *(iii)* Healthcare Policy: integrating metadata from DCAT-compliant nanopublications with national health metadata catalogues to support the European Health Data Space; and *(iv)* Biotech Provenance: FAIRifying mass spectrometry workflows by providing decentralized infrastructure for tracking provenance across heterogeneous digital objects.

More use cases can be found at <https://w3id.org/spaces/knowledgepixels/incubator>, and we are actively seeking new partners to join the program and explore how the platform can be applied to address their specific knowledge-sharing challenges.

4. Impact

The adoption of the nanopublication-based platform within the use cases described in Section 3 illustrates the potential for a significant shift in how knowledge is represented, governed, and accessed. The impact can be evaluated across three main perspectives:

- **Capturing “Off-the-Record” Knowledge:** In traditional ecosystems, valuable information such as peer feedback, informal opinions, or negative results are often lost in volatile communication channels. Nanopublications lower the technical barrier by providing a lightweight, structured, and citable format for recording and disseminating such information. We acknowledge, however, that the mere availability of a publishing infrastructure does not by itself resolve the underlying documentation process challenge: researchers and practitioners must still be motivated and supported in capturing such knowledge. The decentralized platform we propose can thus serve as an enabling substrate for turning “off-the-record” interactions into permanent and actionable knowledge, provided that appropriate documentation practices are in place.
- **AI-Readiness:** Large Language Models (LLMs) could leverage the high-precision data provided to improve their reasoning capabilities and reduce hallucination. The use cases demonstrate that the nanopublication ecosystem provides the high-precision, structured data required for the

next generation of AI tools. By treating AI agents as first-class citizens that can both query and publish nanopublications, the platform reduces the risk of hallucination and enables automated, verifiable reasoning. The information is signed and immutable, ensuring data is trustworthy and that traceability and provenance are maintained throughout.

- **Decoupling Knowledge from Infrastructure:** The P2P nature of the platform ensures that knowledge is not siloed within specific services or institutions, but rather exists as a globally accessible and interoperable graph. Whether it is chemical risk data or health metadata, the information remains globally integrated and locally owned, satisfying the “Interoperability” and “Reusability” pillars of the FAIR principles without requiring a central authority. This decoupling is a core design property of the platform: it ensures long-term sustainability of the broader knowledge ecosystem independently of any single institution or infrastructure provider.

5. Conclusion and Future Work

This paper presented the nanopublication-based ecosystem as a decentralized infrastructure for sharing knowledge in a trusted and machine-actionable way. Rather than positioning nanopublications as a solution in themselves, we argue that they serve as a foundational enabler: a lightweight, verifiable, and interoperable building block upon which a global knowledge commons can be constructed. By exploiting the nanopublication-based platform across the use cases mentioned in Section 3, we have shown that it can support FAIR-compliant knowledge sharing across heterogeneous domains without requiring a central authority.

In future work, we will continue to grow the nanopublication ecosystem and lower the barriers to adoption for a broader range of researchers and practitioners. More specifically, we will investigate the role of AI agents as first-class citizens in the platform, capable of both querying and contributing to the knowledge commons in a verifiable and accountable manner.

Declaration on Generative AI

During the preparation of this work, the author(s) used Gemini 3 in order to: Grammar and spelling check, Improve writing style, Drafting content. After using these tool(s)/service(s), the author(s) reviewed and edited the content as needed and take(s) full responsibility for the publication’s content.

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